Solving Quadratics by Factorising

Exam code: 0607

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Updated on 19 January 2026

Solving quadratics by factorising

How do I solve a quadratic equation using factorisation?

- **Rearrange** it into the form $ax^2 + bx + c = 0$
  - Zero must be on one side
  - It is easier if you rearrange so that a is positive
- **Factorise** the quadratic and **solve** each bracket equal to zero
  - If $(x + 4)(x - 1) = 0$, then either $x + 4 = 0$ or $x - 1 = 0$
    - Because if two things multiply together to give zero,
      - then one or the other of them must be equal to zero
- To solve $(x - 3)(x + 7) = 0$
  - ...solve **first bracket = 0**:
    - $x - 3 = 0$
    - add 3 to both sides: $x = 3$
  - ...and solve **second bracket = 0**
    - $x + 7 = 0$
    - subtract 7 from both sides: $x = -7$
  - The **two solutions** are $x = 3$ or $x = -7$
    - The solutions in this example are the numbers in the brackets, but with opposite signs

What if there are numbers in front of the x's in the brackets?

- The process is **the same**
  - There's a bit more work to find the solutions
  - You can't just write down the answers by changing the signs
- To solve $(2x - 3)(3x + 5) = 0$
  - ...solve **first bracket = 0**
    - $2x - 3 = 0$
    - add 3 to both sides: $2x = 3$

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

        - $3x + 5 = 0$
        - subtract 5 from both sides: $3x = -5$
        - divide both sides by 3: $x = -\frac{5}{3}$
    - The **two solutions** are $x = \frac{3}{2}$ or $x = -\frac{5}{3}$

What if x is a factor?

- The process is **the same**
    - Just be sure to handle the x correctly
    - That 'x as a factor' gives one of the solutions
- To solve $x(x-4) = 0$
    - it may help to think of x as $(x - 0)$ or (x)
    - ...solve **first bracket = 0**
        - $(x) = 0$, so $x = 0$
    - ...solve **second bracket = 0**
        - $x - 4 = 0$
        - add 4 to both sides: $x = 4$
    - The **two solutions** are $x = 0$ or $x = 4$
- It is a common **mistake** to divide (cancel) both sides by x at the beginning
    - If you do this you will **lose a solution** (the $x = 0$ solution)

Examiner Tips and Tricks

Your calculator might be able to solve quadratics. If it does, then you can use this to help you to factorise the quadratic.

For example, a calculator gives solutions to $6x^2 + x - 2 = 0$ as $x = -\frac{2}{3}$ and $x = \frac{1}{2}$. You can rearrange these solutions to get zero on one side, $3x + 2 = 0$ and $2x - 1 = 0$. The left-hand side expressions are two factors, $6x^2 + x - 2 = (3x + 2)(2x - 1)$.

Worked Example

(a) Solve $x^2 + 3x - 10 = 0$ by factorising.

Answer:

Factorise the left-hand side

$$(x - 2)(x + 5) = 0$$

Set the first bracket equal to zero

$$x - 2 = 0$$

Add 2 to both sides

$$x + 5 = 0$$

Subtract 5 from both sides

$$x = -5$$

Write both solutions together using "or"

$$x = 2 \text{ or } x = -5$$

(b) Solve $5x^2 - x = 0$ by factorising.

Answer:

Factorise the left-hand side

$$x(5x - 1) = 0$$

Do not divide both sides by x (this will lose a solution at the end)
Set the first bracket equal to zero

$$(x) = 0$$

Solve this equation to find x

$$x = 0$$

Set the second bracket equal to zero

$$5x - 1 = 0$$

Add 1 to both sides

$$5x = 1$$

Divide both sides by 5

$$x = \frac{1}{5}$$

Write both solutions together using "or"

$$x = 0 \text{ or } x = \frac{1}{5}$$

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